

Transported cell-composition corrections of epigenetic age can add the confounding they are meant to remove

Estimation noise, model shift, and the reach of a penalty

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Abstract Epigenetic age acceleration in blood is routinely adjusted for immune cell composition by regressing clock age on estimated cell proportions. Within the cohort where it is fitted the adjustment does what it claims; we study what happens when its coefficients are applied to another cohort. In six public blood cohorts, scoring only clocks never trained on the cohorts involved, a correction fitted on forty samples and transported increased the composition signal in 83–95% of draws for the age-estimating clocks; for the pace-of-ageing clock DunedinPACE it was neutral. In the units a study reports, it moved the estimated effect of smoking or disease 0.91 years away from the within-cohort estimate, against 0.55 years for applying no correction at all, and reversed its sign in 3 of 24 configurations. The error has two parts: estimation noise, which falls as $1/n$ and is reproduced by coefficients that carry no information, and model shift — a composition effect differing between cohorts — which did not fall with fitting size up to 2,639 samples. Because model shift depends on the target’s own coefficients, nothing computable beforehand certified a transport as safe: of 73 transports a closed-form predictor called safe, 30 were harmful. A fixed ridge penalty cut harmful (pair $\times$ clock) transports from 23 of 72 to 1, where a penalty fading with sample size, or one chosen by cross-validation, did not. In saliva, where composition accounts for up to half of age acceleration, transports were harmful in 7 of 8 cells with or without the penalty, and pooling studies did not help. What separates the tissues is how far cohorts disagree relative to the average effect: the between-cohort standard deviation of the composition slope was 0.09–0.55 of the mean effect in blood and 0.91–5.34 in saliva. A penalty shrinks a coefficient toward zero, which is close to every cohort’s own only in the first case.

1 Introduction

Blood composition shifts with age: lymphocytes decline, myeloid cells increase, and memory cells replace naive ones. Each leukocyte subtype has its own methylation profile, so a clock applied to whole blood partly measures the composition of the sample¹. With twelve cell types resolved, composition explains 13–34% of age-acceleration variance depending on the clock², and naive CD8 T cells read 15–20 years younger than effector memory CD8 cells from the same donor³.

The usual remedy is to regress clock age on chronological age and estimated proportions, and keep the residual. This is normally done inside the dataset at hand, where it removes the linear composition term exactly. For the same reason it cannot be checked there: a least-squares residual is orthogonal to its predictors regardless of how good the coefficients are. Published evaluations of cell-type adjustment are simulation-based and within-dataset⁴.

We examine the case where the coefficients leave the cohort that produced them — reuse of published coefficients, small cohorts borrowing from larger ones, or a fixed correction applied to new samples. Adjustment coefficients fitted in one dataset and applied to others are already published practice: a saliva adaptation of blood clocks fits them on eight pooled studies and applies them to held-out ones⁵, and its training set includes GSE78874, one of the cohorts where we find transport fails. We test it in blood, and in saliva, where composition is larger and a published adaptation already transports it.

2 Methods

2.1 Cohorts, reference panels and clocks

Four public whole-blood 450k series with chronological age: GSE40279 ($n = 656$, ages 19–101), GSE61151 ($n = 184$), GSE50660 ($n = 464$, smoking cohort) and GSE42861 ($n = 689$, rheumatoid arthritis case-control). Proportions were estimated by constrained non-negative least squares against a six-type panel (GSE35069) and a twelve-type panel resolving naive and memory lymphocytes (GSE167998)⁶, both built here; the twelve-type panel recovers known mixture proportions at $r = 0.79$, mean absolute error 0.027. Because both were built here and share their construction, every blood result was repeated on published libraries: the Salas et al. 2022 twelve-type reference on its IDOL-optimised 450k probe set for the fit⁶ and the published seven-type blood reference for the measurement⁷, which share 20 probes of 600 and 333.

Clocks: Horvath 2013⁸, Levine 2018⁹ and Horvath 2018¹⁰. A clock is excluded from any pair involving a cohort it was trained on. Hannum 2013 and Horvath 2013 were both trained on GSE40279 (Horvath 2013 lists it as training set 3; GSE42861 was a test set only). Levine 2018 (InCHIANTI) and Horvath 2018 were trained on none of the four. Pairs involving GSE40279 are therefore scored with Levine 2018 and Horvath 2018 only. A fifth cohort, GSE132203 ($n = 795$, EPIC array, mostly African American), tests replication across array generation and ancestry; clocks are scored on it only above 95% probe coverage (Levine 2018, Horvath 2018). As a clock of a different kind we add DunedinPACE¹¹, which estimates the pace of ageing and was trained on a cohort not in GEO; we reimplemented it from its package’s published model data and validated it (cohort means 0.93–1.05; current smokers +0.14 faster than never smokers, $p = 3 \times 10^{-11}$). A sixth cohort, GSE55763¹² (450k, London), serves as a large fitting cohort: after dropping all 72 technical-replicate arrays it has 2,639 unrelated adults (ages 24–75); it postdates Horvath 2013 and is in no clock’s training set, and every clock cleared the coverage and age checks on it.

Saliva, a mixture of buccal epithelium and leukocytes, was tested in three adult cohorts: GSE232891 (EPIC, $n = 552$; inflammatory bowel disease and controls), GSE232332 (EPIC, $n = 265$ after removing technical replicates; oesophageal cancer and controls) and GSE78874 (450k, $n = 259$; betas from raw signal). Proportions came from the EpiDISH references^{7,13}: a nine-type hierarchical fit (epithelium, fibroblast, seven immune subtypes) and a three-type measurement (epithelium, fibroblast, immune), which share their first step and so bias measurement toward the correction. The first two cohorts come from one group and their files carry no genotyping probes, so shared individuals could not be excluded and they were never paired. Levine 2018 and Horvath 2018 cleared coverage and age checks in all three. Because that measurement is the fit’s own first step, saliva was also scored with an independent panel built from GSE147318¹⁴, children’s saliva sorted into immune and epithelial fractions; its 300 probes share 3.3% with EpiDISH and it is used as a relative immune score, since its absolute scale does not transfer between studies. A fourth saliva cohort, GSE149747 (EPIC, another group), contributes its 44 baseline samples to the slope comparison and to the pooled fits, but is too small for a transport curve.

2.2 Correction and scoring

On the fitting cohort we fit $y = \beta_0 + \beta_1 a + C\beta_c$ (clock age y , chronological age a , proportions C with one column dropped) and apply $y - (C_{\text{test}} - \bar{C}_{\text{fit}})\beta_c$ in the test cohort. The outcome is the composition term remaining in the test cohort’s age residual – the R^2 increment from composition over chronological age, minus a permutation null, as a share of uncorrected age-acceleration variance. Net damage Δ is this share after correction minus before; positive Δ means the transported correction did worse than no correction. The correction is fitted with twelve types and scored with six to avoid scoring it on its own representation; sensitivity analyses score it on twelve types and on the naive/memory columns alone.

All counts are per clock: a pair where one clock is harmed and another helped counts as two cells. Pooling clocks within a pair concealed a substantial part of the harm in an earlier version of this analysis.

2.3 Two components of transport error

Let b be fitted on cohort A and S_B be the composition covariance of cohort B, both after removing intercept and age. The composition left in B is $(\beta_B - b)' S_B (\beta_B - b)$. With $b = \beta_A + e$ and $\text{Cov}(e) = (\sigma^2/n) S_A^{-1}$, its expectation is a specification term $(\beta_A - \beta_B)' S_B (\beta_A - \beta_B)$ plus an estimation term $\sigma^2 \cdot \text{tr}(S_A^{-1} S_B)/n$. We

call $\text{tr}(S_A^{-1} S_B)/n$ the transport index. The estimation term is the standard excess risk of least squares under covariate shift¹⁵; the specification term is model shift¹⁶. We claim neither as new.

A reference with no information is obtained by shuffling whole rows of the fitting cohort's composition matrix before fitting, which preserves the collinearity between cell types. It is measured at every fitting size.

2.4 Penalty, subsampling and inference

Ridge coefficients are fitted on standardized, age-residualized composition with penalty α times the mean eigenvalue, so that α is comparable across cohorts and sizes; $\alpha = 0$ reproduces least squares. Under distribution shift the optimal penalty need not even be positive¹⁷; a fixed positive penalty is used here as a conservative default. The value $\alpha = 3$ was fixed on the first four cohorts, before GSE132203 and GSE55763 entered the project, so its behaviour on those two and on saliva is out-of-sample for the choice. Subsamples are stratified by age decile. Where configurations share cohorts, significance is assessed by block permutation¹⁸, swapping whole index profiles between cohort pairs.

3 Results

3.1 A transported correction can be worse than none

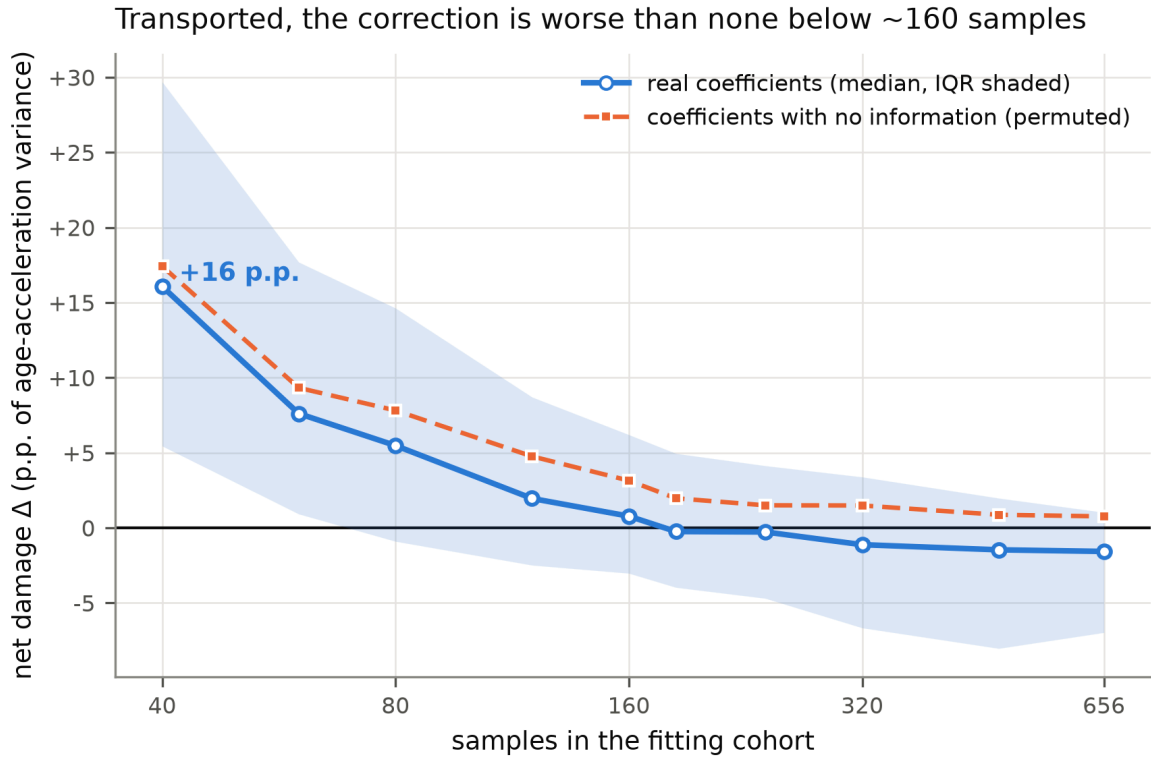


Figure 1. Net damage of a transported correction by fitting size. Fitted on age-stratified subsamples of GSE40279 and applied to three external cohorts; Levine 2018 and Horvath 2018; 100 draws per size. Above zero the correction did worse than none. Orange: the same procedure with composition rows shuffled before fitting.

Fitted on 40 samples and transported, the correction had a median Δ of +16.1 p.p. (IQR +5.5 to +29.7) and was harmful in 88% of draws over 100 draws per size (Figure 1); three earlier sets of 30 draws gave medians from +11.8 to +18.9, so single small sets are unstable at this size. The median was near zero between 160 and 240 samples and -1.6 p.p. at the full 656, where 33% of (clock $\times$ cohort) values were still harmful. Scored on twelve cell types instead of six, the damage at $n = 40$ was +48.1 p.p.; scored on the naive/memory columns alone, +30.6. That measurement shares the fitting panel and is biased toward the correction, so the six-type curve understates the harm.

3.2 Two components

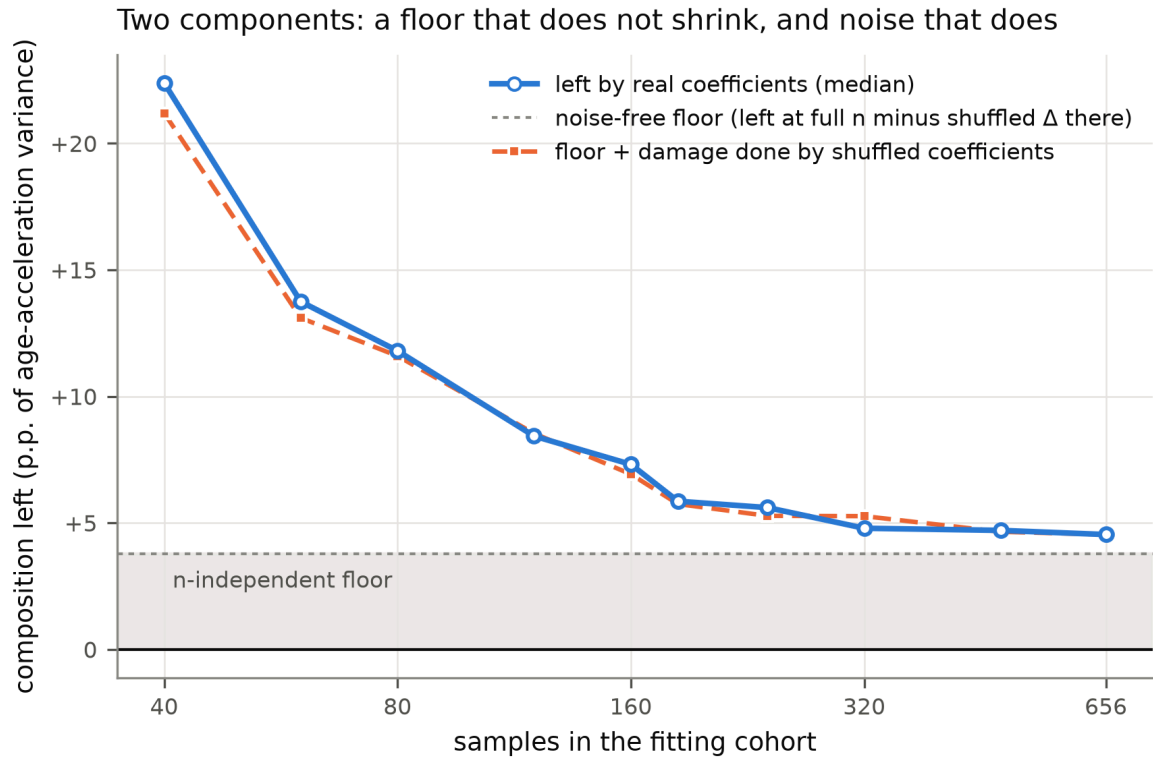


Figure 2. Composition left by the real fit, against a floor plus noise. Blue: what the transported correction leaves. Grey: the noise-free floor, the 3.8 p.p. left at full fitting size after subtracting what shuffled coefficients still do there. Orange: that floor plus the damage shuffled coefficients do at each size. Both series are composition left, in the same units.

Shuffled coefficients carry no information, yet at $n = 40$ they did +17.4 p.p. of net damage, falling with n at a log-log slope of -1.16 (Figure 2). Most of the small-sample harm is therefore estimation noise. The real fit, however, still left 4.5 p.p. of composition at $n = 656$, where shuffled coefficients still did 0.8; the difference, 3.8 p.p., is a floor that fitting size does not remove. Adding that floor to the shuffled damage reproduces the real fit's curve within 0.6 p.p. at every size from 60 upward, and underestimates it by 1.2 at $n = 40$ (Figure 2). The floor is not an artefact of scoring with a different panel: applied within GSE40279, the same correction removed 95% (Levine) and 85% (Horvath 2018) of the six-type composition signal. Transported at full size, it removed between 92% and -109% of the signal it found, depending on the pair – in the worst case doubling it. (Percentages of the signal found are relative; Δ and composition left are in p.p. of age-acceleration variance throughout.)

3.3 Model shift

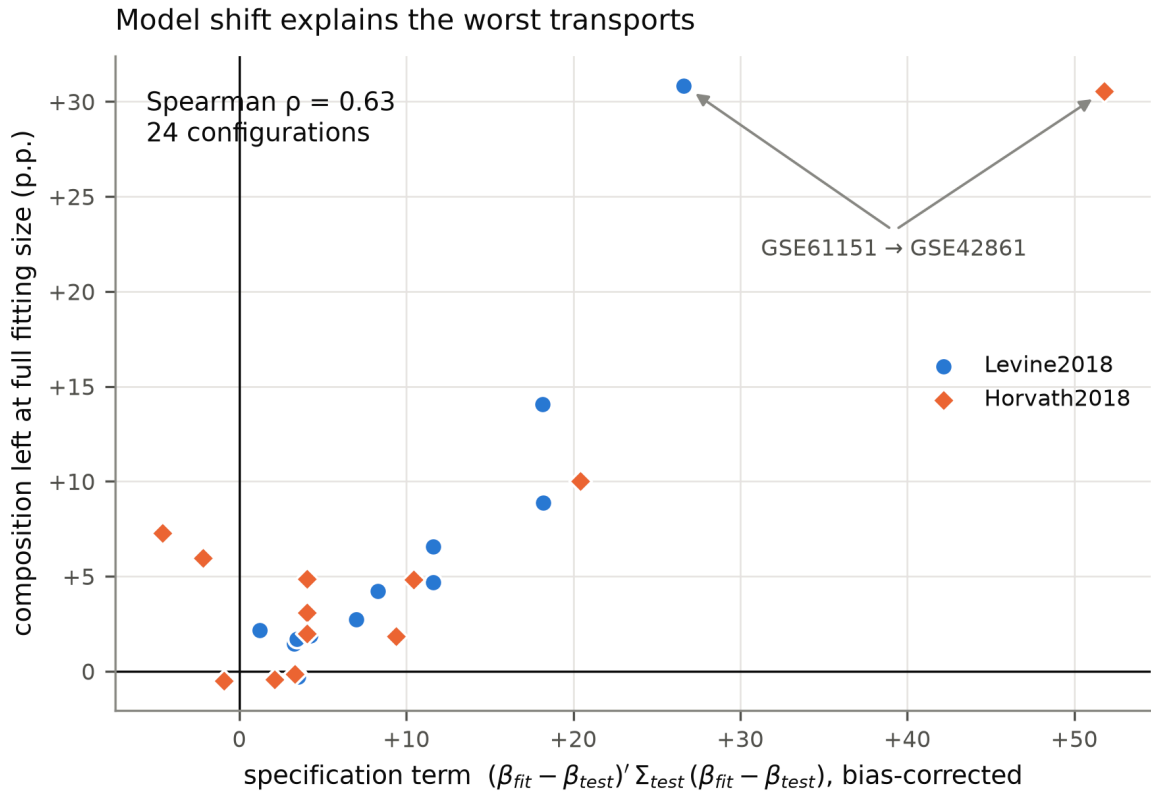


Figure 3. Specification term against composition left at full fitting size. 24 (directed pair $\times$ clock) configurations. The term is bias-corrected for the estimation noise of both fits.

The bias-corrected specification term ranked the full-size leftover at Spearman $\rho = 0.633$ ($p = 0.0009$; Figure 3). It accounts for the worst transport at full fitting size, GSE61151 into the arthritis cohort: predicted +26.6 and +51.8 p.p. for the two clocks, observed +30.8 and +30.5. Pairwise Wald tests of equal coefficients rejected in 4 of 12 comparisons after Bonferroni correction, short of our pre-set threshold of 6 (9 of 12 at nominal $p < 0.05$). The two measures weight coefficient differences differently: Wald by estimation precision, the specification term by variance in the target cohort. Within cohorts, Levine 2018 coefficients differed between arthritis cases and controls and between ever and never smokers ($p = 0.013$ each); for Horvath 2018 this test did not reject ($p = 0.61, 0.48$). Adjusting for disease and smoking left between-cohort differences essentially unchanged.

Model shift is not a batch effect. Within GSE42861, where cases and controls share study, array and laboratory, we fitted the correction on a random half of the controls and applied it both to the other half and to the arthritis cases (30 paired splits). Applied to cases it left 7.4, 7.4 and 2.6 more points of composition than applied to controls (Horvath 2013, Levine 2018, Horvath 2018), although the transport index was lower for cases — so none of the excess is estimation noise, and the estimated model-shift component was 8.7, 8.4 and 6.3 points. Horvath 2018 is affected even though the Wald test above did not detect a coefficient difference. The choice of reference population also changes the estimated disease effect: for Horvath 2018, the age-adjusted arthritis effect was -0.74 years (95% CI -1.05 to -0.37) with a whole-cohort correction and -1.39 (-2.08 to -0.80) with a controls-only one, a difference of -0.66 (-1.13 to -0.31) by a bootstrap that refits both corrections in each resample.

3.4 What can be known before transporting

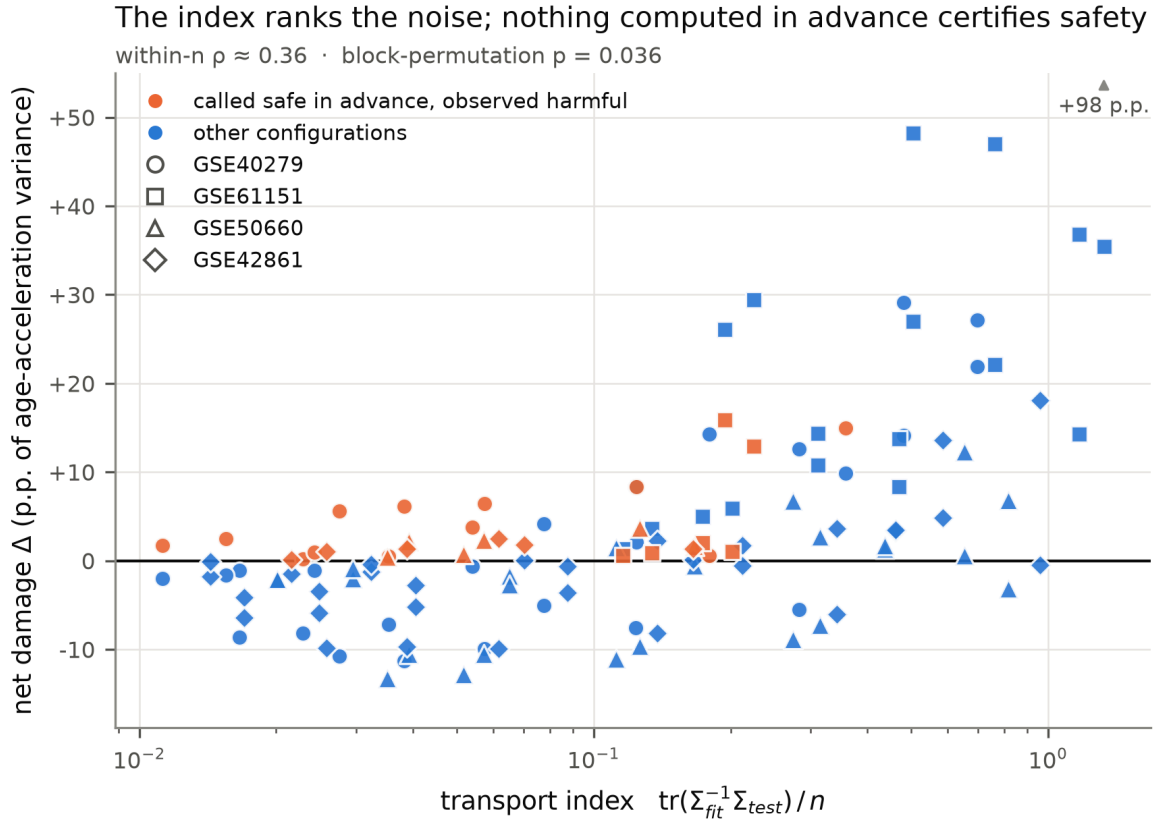


Figure 4. Transport index against net damage, per clock. Orange: transports that the closed-form predictor labelled safe and that were harmful. Marker shape gives the fitting cohort.

Because the index scales as $1/n$, part of its correlation with damage comes from sample size alone. At the three sizes available for every pair, a block permutation that keeps each pair’s fitting sizes gave a null median of $\rho = 0.42$; the observed value was 0.579 ($p = 0.036$). Within a single fitting size the index ranked damage at $\rho = 0.41, 0.36$ and 0.33 . It carries real but modest information about which pairs are risky (Figure 4). Counted per clock, 12 of 40 transports with an index below 0.05 were harmful, so no index threshold marks a safe region.

A closed-form estimate of net damage that assumes shared coefficients, $2\hat{\sigma}^2 \cdot \text{index} - b'S_B b$, needs only the fitting cohort and the target’s proportions. It predicted the sign in 71% of configurations ($\rho = 0.6332$ over 126 configurations — coincidentally near the specification term’s ρ above, which is a different correlation over 24), against 81% ($\rho = 0.884$) for an oracle that knows the target’s own coefficients. It errs toward reassurance: 30 of the 73 transports it labelled safe were harmful, concentrated where the specification term is large.

3.5 A penalty removes most of the harm

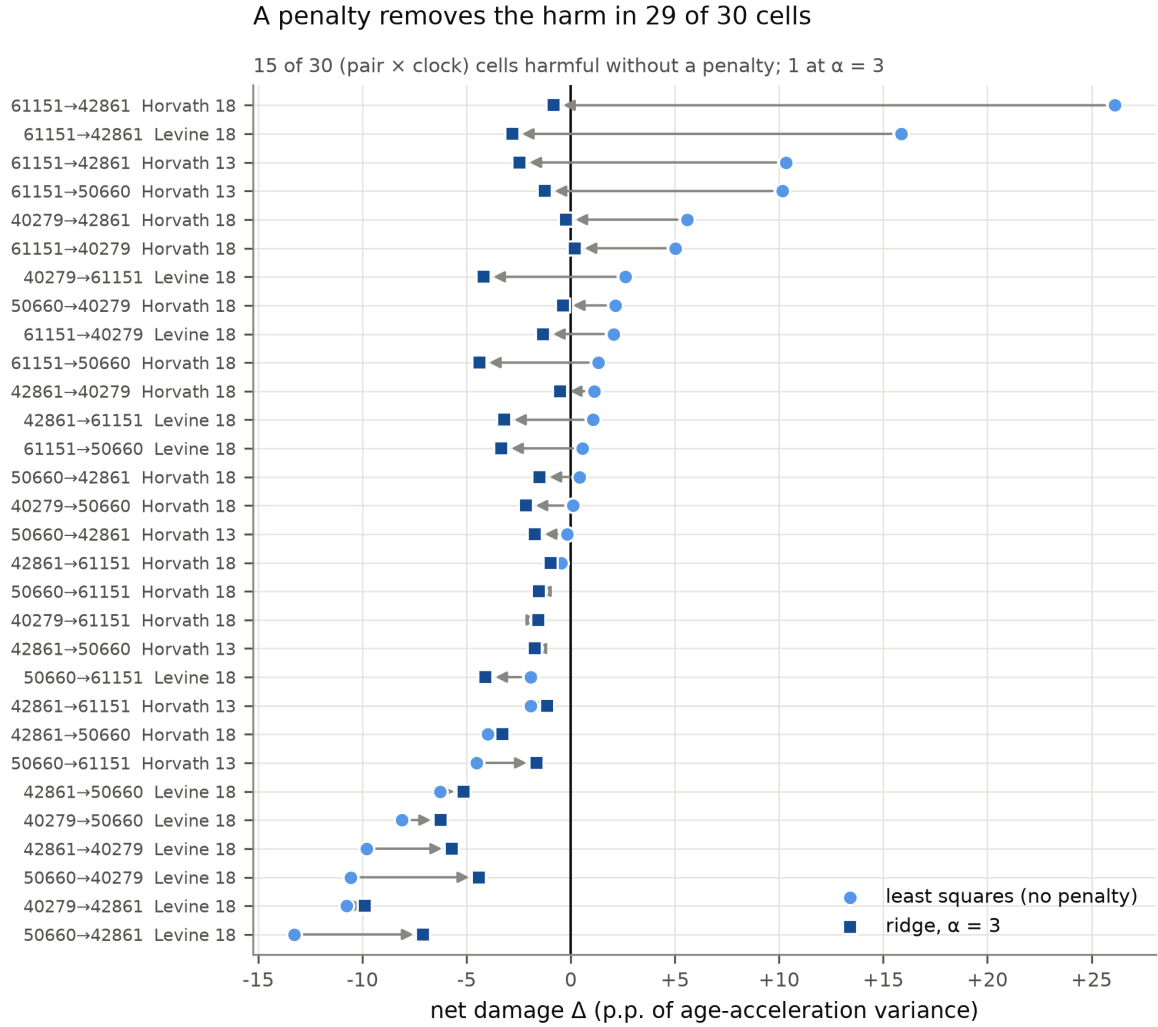


Figure 5. Unpenalised against ridge-penalised transport, per (pair $\times$ clock) cell at matched fitting size $n = \min(n_A, n_B)$.

Without a penalty, 15 of 30 (pair $\times$ clock) cells were harmful. At $\alpha = 3$, one remained, at +0.2 p.p. (Figure 5); at $\alpha = 10$, none. The penalty has a cost: where the unpenalised correction already helped, the median benefit fell from -4.0 p.p. to -3.3 at $\alpha = 3$ and to -1.5 at $\alpha = 10$. The fixed penalty also held when the two cohorts were deconvolved from different reference panels (14 harmful cells unpenalised, 1 at $\alpha = 3$) and when scored on twelve types or the naive/memory axis (no harmful cell at $\alpha = 3$).

Choosing the penalty by leave-one-out cross-validation on the fitting cohort did not work as well: it selected $\alpha = 0.3$ in the median cell and left 30% of cells harmful, against 3% at fixed $\alpha = 3$. Cross-validation optimises fit within the fitting cohort and cannot account for where the coefficients will be used.

The penalty's size is defined relative to the fitting cohort's composition variance, so its proportional shrinkage does not fade with n . A conventional fixed- λ penalty does fade; set equal to $\alpha = 3$ at $n = 40$, it is 0.045 at $n = 2639$. Over all 30 directed pairs among the six cohorts at matched size (72 age-clock cells), it left 11 cells harmful against 1 for $\alpha = 3$ (23 unpenalised); the cells it missed were the model-shift transports. This matches the distinction between covariate and regression shift in optimal ridge regularisation¹⁷: an error that does not shrink with n is not bounded by a penalty that does.

3.6 A pace-of-ageing clock

For DunedinPACE, the transported correction fitted on 40 samples of GSE40279 was neutral (median +0.3 p.p., harmful in 51% of draws), although shuffled coefficients still did +4.9 p.p. of damage: the noise component was present, but real coefficients removed enough genuine signal to offset it. Fitted instead on the fifth cohort, it

was neutral again (−1.5 p.p., 42%), so this looks like a property of the clock rather than of the fitting cohort. What replicated was the rest: at matched sizes, 6 of 12 directed pairs were harmful unpenalised and none at $\alpha = 3$, and a correction fitted on controls left 10.5 more points of composition in arthritis cases than in other controls.

3.7 Replication on another array and ancestry

Fitted on 40 samples of GSE132203 (EPIC, mostly African American) and transported to the four 450k cohorts, the correction was harmful for the age clocks in 95% of draws (median +24.3 p.p.; Levine 2018 91%, Horvath 2018 99%), with shuffled coefficients at +16.4. Across the 8 directed pairs involving this cohort, 7 of 24 (pair × clock) cells were harmful unpenalised and none at $\alpha = 3$; the 24 count DunedinPACE alongside the two age clocks (6 of its 16 age-clock cells were harmful).

3.8 A fitting cohort four times larger

Fitted on GSE55763 and transported to the other five cohorts (13 age-clock cells), the correction was harmful at $n = 40$ in 83% of draws (median +7.6 p.p.). Between $n = 656$ and the full 2,639 the composition it left fell only from 1.9 to 1.3 points, a ratio of 0.72 against the 0.25 that $1/n$ noise predicts; the shuffled reference fell from 0.5 to 0.1 (log-log slope −1.03). The floor therefore persists on median. It is not uniform: at full size it was +4.6 points into the arthritis cohort and +3.0 into GSE40279, but zero into GSE50660 and GSE61151, and it exceeded the shuffled reference in 8 of 13 cells, below our pre-set 9. Model shift is a property of the pair of cohorts. With this many fitting samples the unpenalised correction helped in 10 of 13 cells, and the fixed penalty cost benefit (median −3.0 p.p. against −4.0) while removing the three harmful cells. Restricting GSE40279 to the fitting cohort’s age range (24–75 years) left its floor unchanged (+3.5 to +3.3 points for Horvath 2018), so age extrapolation does not explain it. The floor is confined to the cohort’s European-ancestry half (+4.3 and +3.9 points, against −0.4 and −0.2 in its Hispanic half), which in this cohort was also processed on separate plates; centring by plate did not remove it.

3.9 Saliva

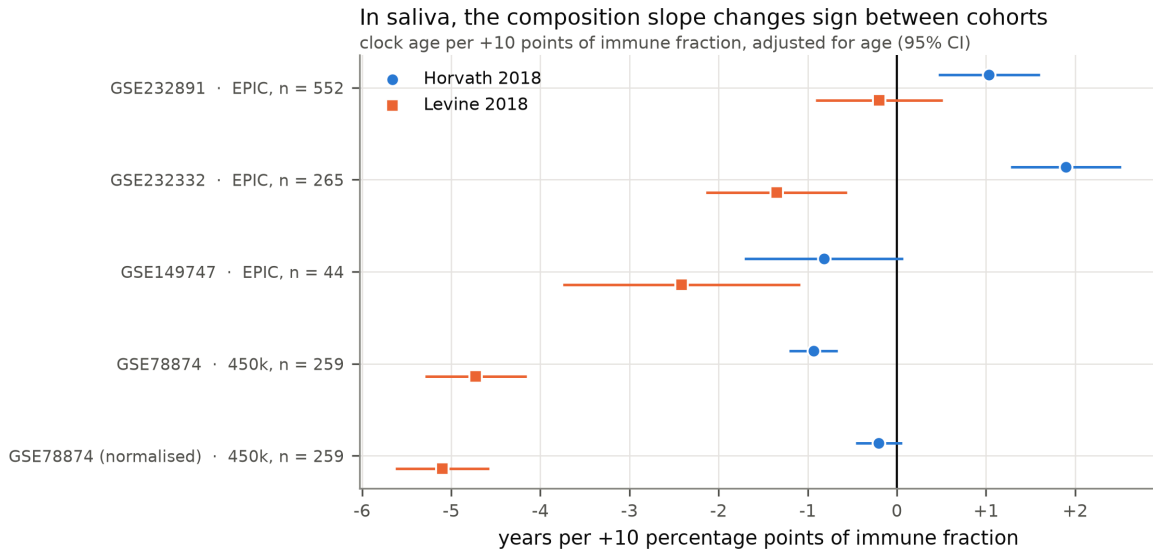


Figure 6. Immune-fraction slope of each clock in four saliva cohorts. Clock age regressed on chronological age and immune fraction; slope per 10 percentage points, 95% CI. GSE149747 at baseline.

Before correction, composition accounted for 10.8–51.6% of age-acceleration variance for the age clocks in saliva, and 42–56% for DunedinPACE. Fitted on 40 samples and transported between saliva cohorts, the correction was harmful in 67% of draws (median +20.9 p.p.; shuffled +6.4). At matched size ($n = 259$), 7 of 8 (pair × clock) cells were harmful unpenalised, 7 of 8 at $\alpha = 3$ and 6 of 8 with the fading penalty; for Horvath 2018 into GSE78874 the unpenalised correction left 132 and 168 p.p. of composition where it had found 16 — eight to ten times the signal it was meant to remove. Within each cohort, fitting on one random half and applying to the other was beneficial in 5 of 6 age-clock cells (−15 to −48 p.p.); the exception had 132 fitting

samples and a small initial signal. The failure is therefore in the transport. Every saliva pair also crosses array and preprocessing, so technical and biological differences cannot be separated, but quantile-normalising GSE78874 to the EPIC distribution left 7 of 8 cells harmful, and so did a three-type fit whose composition matrix is well conditioned (condition number ≈ 1 , against 872–1,317 for nine types). The cause is on the dominant axis (Figure 6): adjusted for age, Horvath 2018 changed by +1.0 and +1.9 years per 10 points of immune fraction in the EPIC cohorts and by -0.9 in GSE78874, and Levine 2018, which keeps its sign in all four cohorts, ranged from -0.2 to -4.7. Shrinkage moves a transported coefficient toward zero, so it can only limit a correction that disagrees with the target this much: at $\alpha = 3$ the worst cell fell from +116 to +28 p.p., and Levine 2018's cells, sign intact, to a median of +3.2 rather than to zero. In a fourth EPIC saliva cohort from another group (GSE149747, 44 adults at baseline) the slope was -0.82 (95% CI -1.71 to 0.07), on the side of the 450k cohort rather than the other EPIC cohorts, which argues against the array as the explanation; by our pre-set rule this comparison was inconclusive.

Pooling studies did not rescue transport: fitted on the other saliva cohorts with a fixed effect per study and applied to the one left out, the correction was harmful in 6 of 8 cells (5 of 8 at $\alpha = 3$), because the pooled slope takes the sign of the pool's majority.

Blood differs in how far the cohorts disagree relative to the effect itself. In a random-effects fit of the per-cohort slopes, the between-cohort standard deviation over the absolute mean effect was 0.09–0.55 in the six blood cohorts (naive CD8 and neutrophil axes, both clocks) and 0.91 for Levine 2018 and 5.34 for Horvath 2018 in saliva; I^2 was 43–84% against 97%. Sign reversal is the extreme of that spread, and it is not the whole of it: with GSE78874 normalised, Horvath 2018's saliva slope is -0.20 (95% CI -0.46 to +0.06), no longer clearly opposite to the EPIC cohorts, and 7 of 8 cells remain harmful; Levine 2018 never reverses and is still harmful at $\alpha = 3$ in 3 of 4 cells.

3.10 What it does to a reported association

Composition left is not what a study reports. For four exposures — smoking (GSE50660), rheumatoid arthritis (GSE42861), inflammatory bowel disease and oesophageal cancer (the two EPIC saliva cohorts) — we re-estimated the exposure coefficient in (clock age) age + exposure under three corrections: none, fitted within the target, and transported. Taking the within-cohort correction as the comparator, a correction fitted on 40 samples elsewhere moved the reported effect a median of 0.91 years, against 0.55 years for applying no correction at all; at $\alpha = 3$, 0.48; at full fitting size, 0.25. Of 24 (target $\times$ source $\times$ clock) cells, 13 moved by more than a year and 3 reversed sign; the largest was 4.19 years, for Levine 2018 carried from GSE132203 into the arthritis cohort, whose within-cohort effect is +0.04 years. At $\alpha = 3$, 6 of 24 still moved by more than a year.

This metric also shows a tail that composition left, bounded above by construction, cannot: at $n = 40$, 1.2% of 720 draws landed more than 10 years from the within-cohort estimate, the worst at 37.6 years. Solved as exact least squares, without dropping near-zero singular values, 10.1% exceeded 10 years, so the tail's size is solver-dependent while its existence is not; it is concentrated in saliva, where the composition matrix is nearly singular. At $\alpha = 3$ no draw exceeded 10 years.

Reliability is not a substitute for this metric. GSE55763 measured 36 people twice; on those 72 arrays, which the fit never saw, the technical ICC(2,1) of age acceleration fell after the within-study correction for all four age clocks (Horvath 2013 0.817 to 0.769), as adjusting for composition lowers reliability elsewhere¹⁹. A correction transported from 40 samples raised it instead (0.842), above the uncorrected value in three clocks. The reason is arithmetic: the correction left the disagreement between a person's two arrays almost unchanged (3.68 to 3.53 for Horvath 2013) and removed between-person variance (18.32 to 13.56), which is what a composition correction is meant to remove. A correction that removes less therefore scores as more reliable, so reliability cannot rank corrections, while the movement of a reported association can.

3.11 Robustness to the reference panels

Repeated on the published libraries, with nothing else changed, the blood results hold: the correction fitted on 40 samples of GSE40279 was harmful in 77% of 600 draws (median +9.1 p.p., against 88% and +16.1 with the panels built here); over the 60 (pair $\times$ clock) cells at matched size, 17 were harmful unpenalised and none at $\alpha = 3$; and fitted on all of GSE55763 the median composition left was +1.9 p.p. The two twelve-type panels

agree on what they both estimate (neutrophils $r = 0.985\text{--}0.997$, naive CD8 $r = 0.852\text{--}0.919$ across the six cohorts).

In saliva, the independent panel correlates $0.951\text{--}0.997$ with the EpiDISH immune fraction and sees almost the same composition signal before correction as EpiDISH’s own immune column (for example $+1.5$ against -0.1 p.p. in GSE232891 for Levine 2018, $+50.6$ against $+50.6$ in GSE78874). What differs is the number of axes, not their source: the three-type measurement sees $+18.5$ and $+31.8$ p.p. in GSE232891 where the immune axis alone sees about zero, so in the EPIC cohorts most of the composition signal in age acceleration lies on the epithelial/fibroblast axes. Scored on the immune axis alone with the independent panel, 6 of 8 cells were harmful unpenalised and 5 of 8 at $\alpha = 3$, against 7 and 7 with the shared measurement.

4 Discussion

Within-cohort composition adjustment cannot be validated inside the cohort; this follows from least squares, not from data. It is also not free: adjusting for immune fractions within a dataset lowers the biological reliability of nearly every clock¹⁹, so the within-cohort correction we use as a comparator is a reference, not a standard of truth. Transported, the adjustment’s small- sample damage is mostly estimation noise — shuffled coefficients do nearly as much of it — while its large-sample damage comes from differences in the composition effect between cohorts. That second component is invisible without the target’s own coefficients, and a target large enough to estimate them could simply be adjusted within itself.

In the units a study reports, the cost of borrowing a correction is about a year on a disease or exposure effect, and more than four years in the worst configuration we found — enough to change what a paper concludes.

For practice this suggests four things. Where the target cohort is large enough, fit the adjustment within it. In blood, where coefficients must be transported, penalise them with a fixed, substantial penalty rather than one chosen by cross-validation on the source cohort or one that fades with n ; with thousands of fitting samples this costs about a point of benefit. In saliva, and plausibly in any tissue where composition dominates, do not transport: no penalty and no pooling made it safe. And treat any pre-transport diagnostic, including the transport index, as a ranking of risk rather than a guarantee. The same caution applies inside a single study: a correction fitted on controls and applied to patients is a transport, and here it changed the estimated disease effect by up to a factor of two.

The penalty’s reach follows from how far cohorts disagree relative to the effect being corrected. Shrinking toward zero replaces a transported coefficient with a smaller one, which is close to every cohort’s own coefficient when the between-cohort spread is a fraction of the average effect, as in blood, and close to none of them when the spread is as large as the effect, as in saliva. A reversal of sign is the extreme case, not a separate mechanism, and we would expect the penalty to fail wherever that ratio approaches one.

Transport is already published practice outside blood: a saliva adaptation of a blood clock fits composition terms on about 960 pooled samples and applies them to held-out studies⁵. It was judged by accuracy against chronological age, which cannot show composition left in acceleration; in our saliva cohorts, pooling did not prevent harm. Composition-dependent heterogeneity within a saliva cohort has also been reported²⁰.

5 Limitations

Six adult whole-blood cohorts, five on the 450k array and one on EPIC, two defined by disease or exposure, and four adult saliva cohorts whose transport pairs all cross array and preprocessing; other tissues and children are untested. The panels built here were checked against published libraries, and the saliva measurement against an independent panel, without changing any conclusion. Medians at small fitting sizes are unstable across independent sets of 30 draws ($+11.8$ to $+18.9$ p.p. at $n = 40$; the pooled figure from 100 draws is $+16.1$), and the small-sample harm itself depends on the clock and the fitting cohort (neutral for DunedinPACE from GSE40279). The penalty value is specific to this panel and these clocks. The decomposition assumes a linear composition effect; squared terms for the four largest components add a median of 0.002 to within-cohort R^2 (significant in 4 of 12 cohort $\times$ clock cells), and a quadratic correction transports no better than the linear one (22 against 23 harmful cells of 60, same median). Sensitivity scoring on twelve types shares the fitting panel. None of this bears on whether epigenetic clocks measure biological ageing; it concerns one correction applied to them.

6 Data and code availability

All series are public (GSE40279, GSE61151, GSE50660, GSE42861, GSE132203, GSE55763, GSE232891, GSE232332, GSE78874, GSE149747, GSE147318, GSE35069, GSE167998); published reference matrices come from the EpiDISH Bioconductor package (version 2.28.0). Analysis code, figure scripts and the stage-by-stage record of the project, including every conclusion that a later stage overturned, are at github.com/KTHimiko/clock-lab (code under the MIT licence; text and figures under CC BY 4.0). Every number in this manuscript is produced by a script there.

7 Declarations

Use of generative AI. This study was carried out with extensive use of an AI system, Claude (Anthropic), through the Claude Code environment. The AI wrote the analysis code, designed and ran the analyses with checks written before each result was read, searched and summarised the literature, and drafted the text and figures. The author set the aims, decided which questions to pursue and which results to report, obtained informal critical readings of earlier drafts and had the corrections they required made, and takes responsibility for the content. The AI is not an author.

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